MSHA Block 2 Revised Expanded Analysis

Selected unseen narratives: 2,000
Existing records reused: 2,000
New model calls: 0
Cause-analysis population: 1,764
Directly supported factors: 221
Contextually supported factors: 647
Expanded factor-analysis population: 868
Cause-only population: 896

Main change: factor evidence is now tiered as directly supported, contextually supported, or unsupported. Expanded factor and prevention analyses use direct plus contextual support. Direct-only results remain the strict sensitivity population.

Processing and Analytical Populations

Measure	Value
Selected narratives	2000
Existing records reused	2000
New Qwen calls	0
Cause-analysis population	1764
Directly supported factors	221
Contextually supported factors	647
Expanded factor-analysis population	868
Cause-only population	896
Excluded/review population	236
Original outputs recovered from raw response	1970
Original outputs using saved-field fallback	30
Prevention mapping repairs	435
Unsupported-factor abstentions	1063

Factor-Support Tier Distribution

Factor support tier	Count	Percent
NO_SUPPORTED_FACTOR	1063	53.1500
CONTEXTUALLY_SUPPORTED	713	35.6500
DIRECTLY_SUPPORTED	224	11.2000

Revised Analysis Status

Analysis status	Count	Percent
VALID_FOR_CAUSE_ANALYSIS_ONLY	896	44.8000
VALID_CONTEXTUAL_FACTOR_ANALYSIS	647	32.3500
EXCLUDED_OR_REVIEW_REQUIRED	236	11.8000
VALID_DIRECT_FACTOR_ANALYSIS	221	11.0500

Factor Support Methods

Factor support method	Count	Percent
MODEL_ABSTAINED	1015	50.7500
HIGH_CONFIDENCE_WITH_EXPLANATORY_EVIDENCE	544	27.2000
TRACEABLE_MODEL_EVIDENCE	199	9.9500
CATEGORY_SPECIFIC_NARRATIVE_CUE	169	8.4500
INSUFFICIENT_FACTOR_SUPPORT	48	2.4000
TRACEABLE_MODEL_EVIDENCE_WITH_CATEGORY_CUE	25	1.2500

Prevention Assignment Methods

Prevention assignment method	Count	Percent
UNSUPPORTED_FACTOR_ABSTENTION	1063	53.1500
ORIGINAL_ALLOWED_PREVENTION	502	25.1000
PREFERRED_MAPPING_REPAIR	435	21.7500

Factor-Evidence Method

DIRECTLY_SUPPORTED: the model evidence is found in the narrative and includes category-specific factor terminology.

CONTEXTUALLY_SUPPORTED: strict evidence matching fails, but the source narrative contains transparent, category-specific contextual cues.

NO_SUPPORTED_FACTOR: no defensible separate factor is retained; prevention is INVESTIGATION_REQUIRED.

The contextual tier is exploratory and will be tested in the 100-record human review.

Primary Cause Distribution

Primary cause	Count	Percent
CAUGHT_IN_OR_BETWEEN	411	23.2993
SLIP_TRIP_OR_FALL	277	15.7029
STRUCK_BY_MOVING_OBJECT	274	15.5329
LOSS_OF_CONTROL_OF_TOOL_OR_EQUIPMENT	197	11.1678
AWKWARD_MOVEMENT_OR_BODY_MOTION	119	6.7460
MANUAL_HANDLING_OR_OVEREXERTION	119	6.7460
FOREIGN_BODY_OR_PARTICLE_CONTACT	87	4.9320
HEAT_OR_ENVIRONMENTAL_EXPOSURE	64	3.6281
OTHER	34	1.9274
GROUND_OR_ROOF_FAILURE	31	1.7574
INSUFFICIENT_INFORMATION	31	1.7574
STRUCK_AGAINST_STATIONARY_OBJECT	29	1.6440
CHEMICAL_DUST_OR_OCCUPATIONAL_EXPOSURE	28	1.5873
PRESSURE_OR_STORED_ENERGY_RELEASE	22	1.2472
ELECTRICAL_CONTACT	18	1.0204
UNEXPECTED_EQUIPMENT_MOVEMENT	12	0.6803
VEHICLE_OR_MOBILE_EQUIPMENT_EVENT	6	0.3401
FIRE_EXPLOSION_OR_THERMAL_EVENT	4	0.2268
ERGONOMIC_OR_REPETITIVE_EXPOSURE	1	0.0567

Expanded Contributing-Factor Distribution

Contributing factor	Count	Percent
UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL	391	45.0461
POOR_HOUSEKEEPING_OR_SURFACE_CONDITION	239	27.5346
UNSTABLE_GROUND_OR_ROOF_CONDITION	57	6.5668
ENVIRONMENTAL_CONDITION	53	6.1060
PROCEDURE_OR_WORK_METHOD	50	5.7604
ENERGY_ISOLATION_DEFICIENCY	46	5.2995
LINE_OF_FIRE_BODY_POSITION	19	2.1889
INADEQUATE_SECUREMENT_OR_RESTRAINT	5	0.5760
PPE_OR_PROTECTIVE_CONTROL	3	0.3456
INADEQUATE_HAZARD_RECOGNITION	2	0.2304
TRAINING_OR_EXPERIENCE	2	0.2304
EQUIPMENT_CONDITION_OR_FAILURE	1	0.1152

Expanded Prevention-Theme Distribution

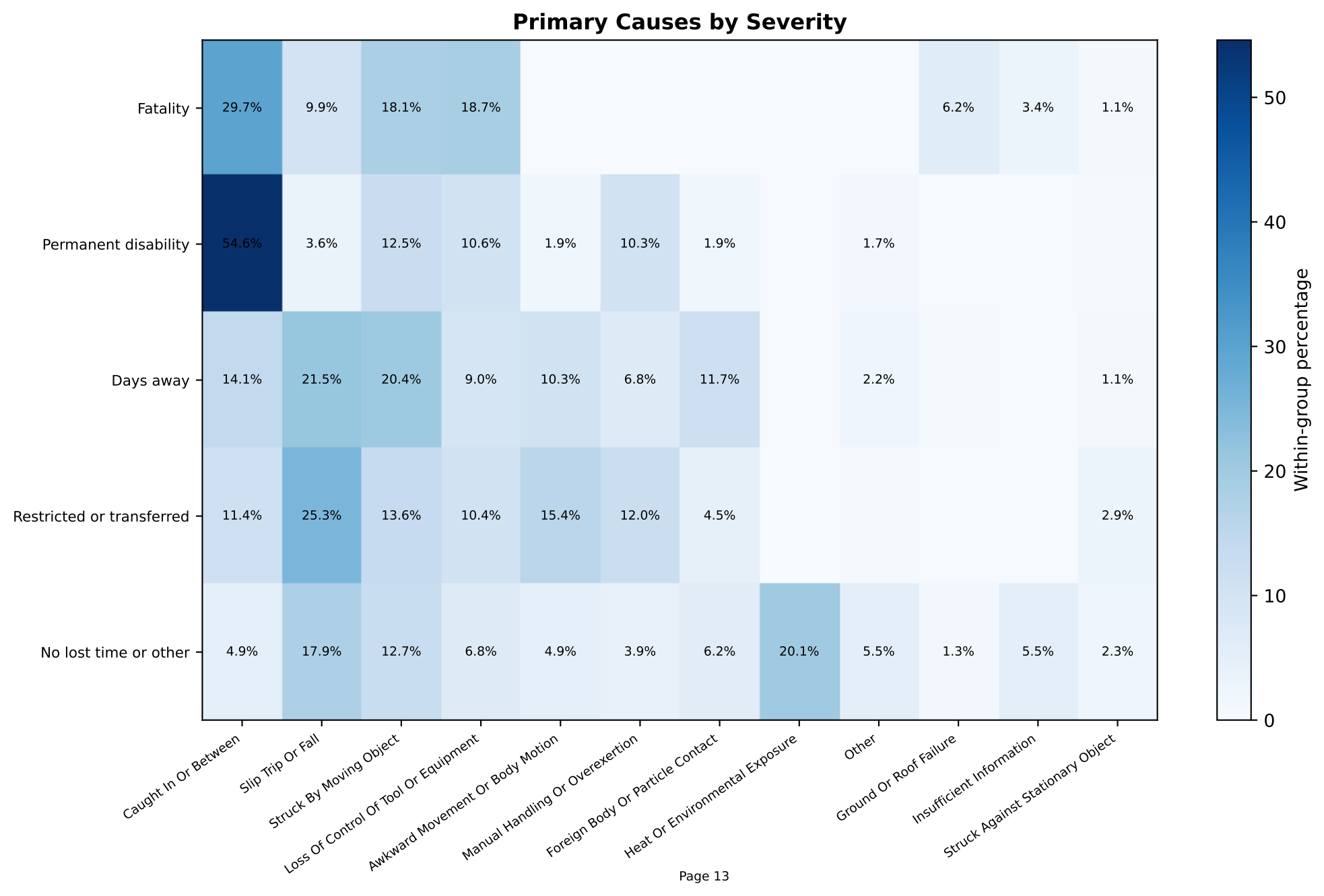
Prevention theme	Count	Percent
SECURE_OR_REPOSITION_MATERIAL	396	45.6221
IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING	239	27.5346
IMPROVE_GROUND_CONTROL_OR_SCALING	57	6.5668
CONTROL_ENVIRONMENTAL_EXPOSURE	53	6.1060
REVISE_PROCEDURE_OR_JOB_PLAN	51	5.8756
ISOLATE_OR_CONTROL_STORED_ENERGY	46	5.2995
ESTABLISH_LINE_OF_FIRE_EXCLUSION	17	1.9585
CONDUCT_PRE_JOB_HAZARD_REVIEW	3	0.3456
IMPROVE_PPE_OR_PROTECTIVE_EQUIPMENT	3	0.3456
PROVIDE_TARGETED_TRAINING	2	0.2304
REPAIR_REPLACE_OR_INSPECT_EQUIPMENT	1	0.1152

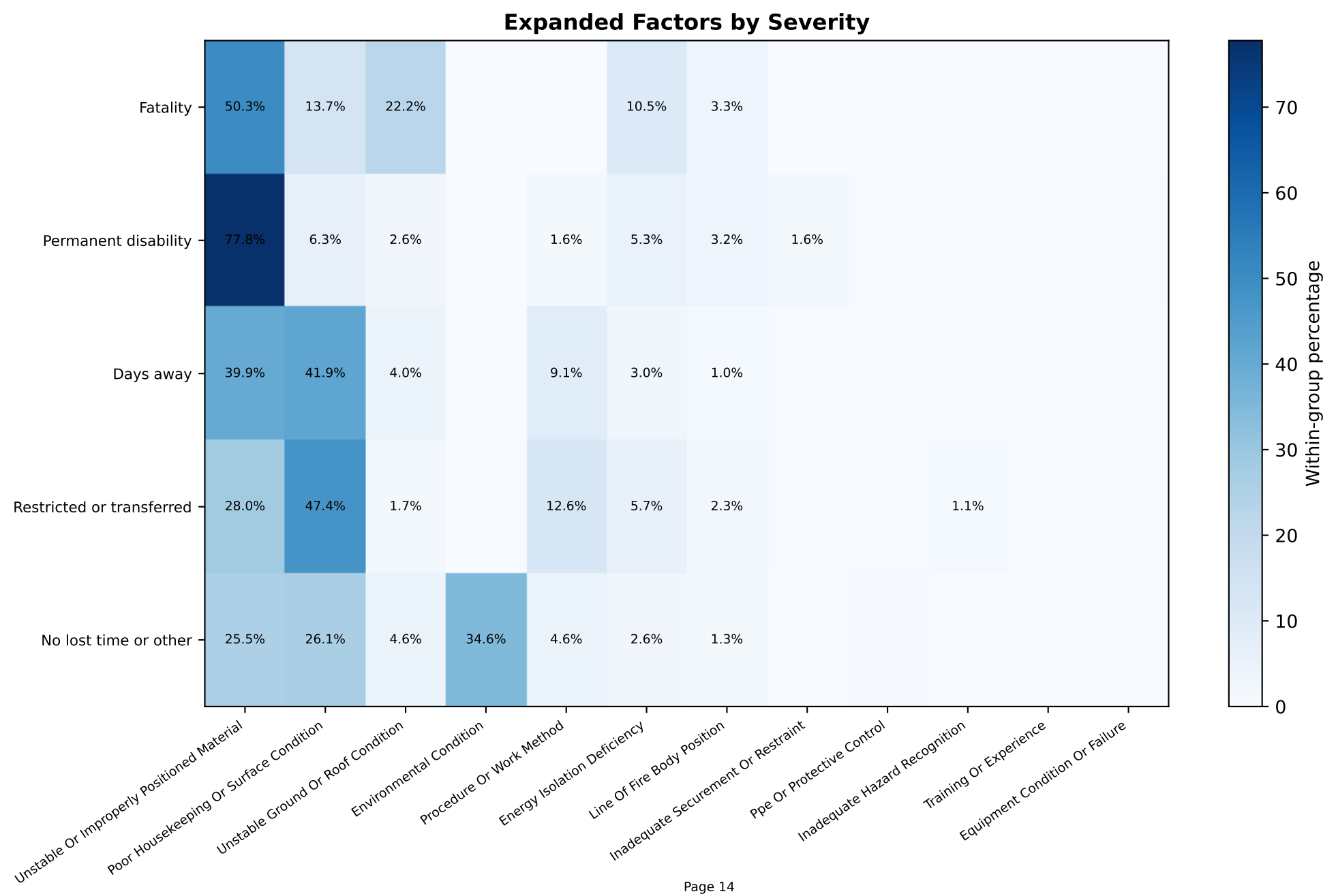
Factor Sensitivity: Direct versus Expanded

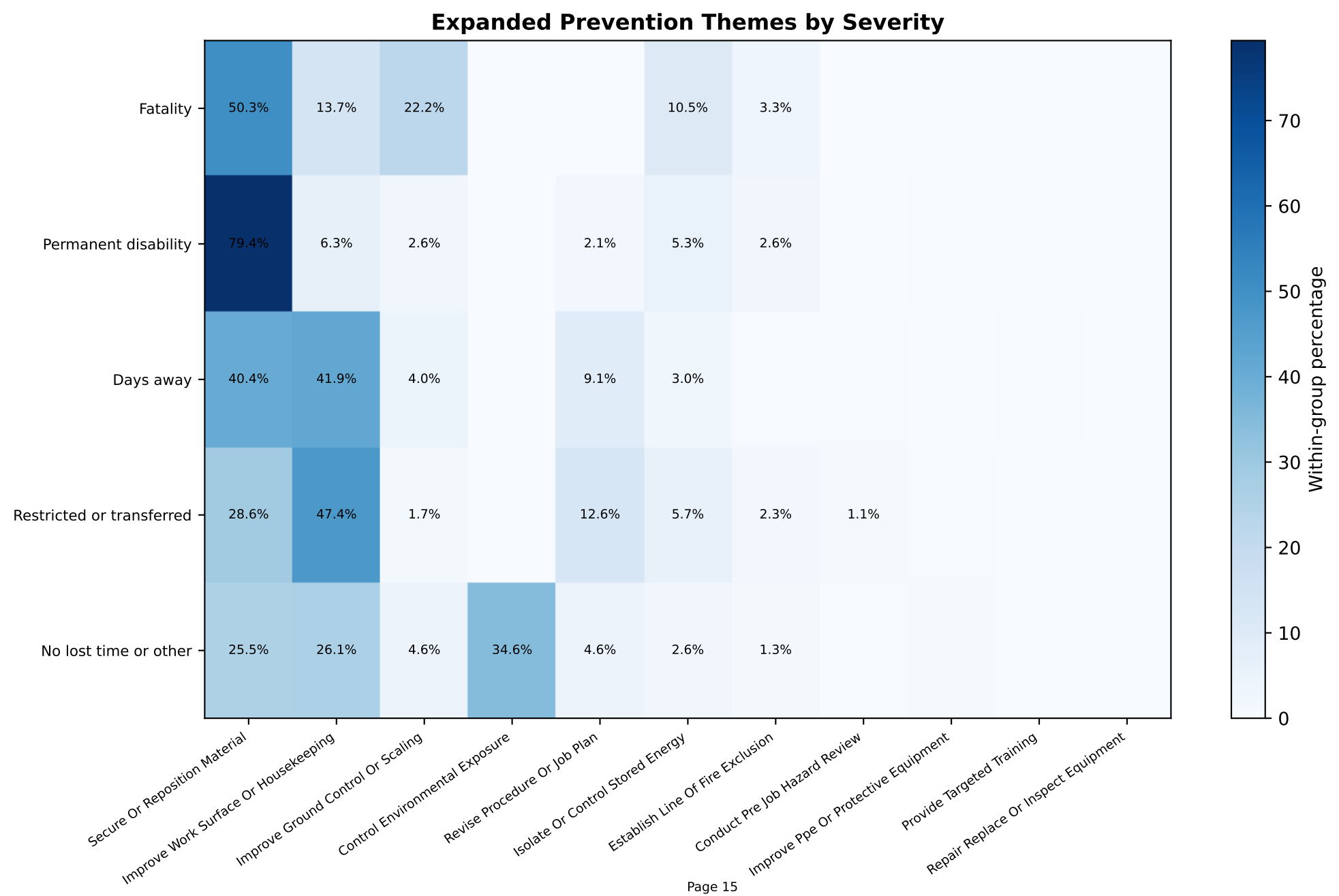
REVISED_FACTOR	Direct support only	Direct plus contextual	Percentage-point difference
ENVIRONMENTAL_CONDITION	0.0000	6.1060	6.1060
UNSTABLE_GROUND_OR_ROOF_CONDITION	4.9774	6.5668	1.5894
PROCEDURE_OR_WORK_METHOD	4.9774	5.7604	0.7830
PPE_OR_PROTECTIVE_CONTROL	0.0000	0.3456	0.3456
INADEQUATE_HAZARD_RECOGNITION	0.0000	0.2304	0.2304
LINE_OF_FIRE_BODY_POSITION	2.2624	2.1889	-0.0735
INADEQUATE_SECUREMENT_OR_RESTRAINT	0.9050	0.5760	-0.3289
EQUIPMENT_CONDITION_OR_FAILURE	0.4525	0.1152	-0.3373
ENERGY_ISOLATION_DEFICIENCY	5.8824	5.2995	-0.5828
TRAINING_OR_EXPERIENCE	0.9050	0.2304	-0.6746
POOR_HOUSEKEEPING_OR_SURFACE_CONDITION	29.8643	27.5346	-2.3297
UNSTABLE_OR_IMPROPERLY_POSITIONED_MATER	49.7738	45.0461	-4.7277

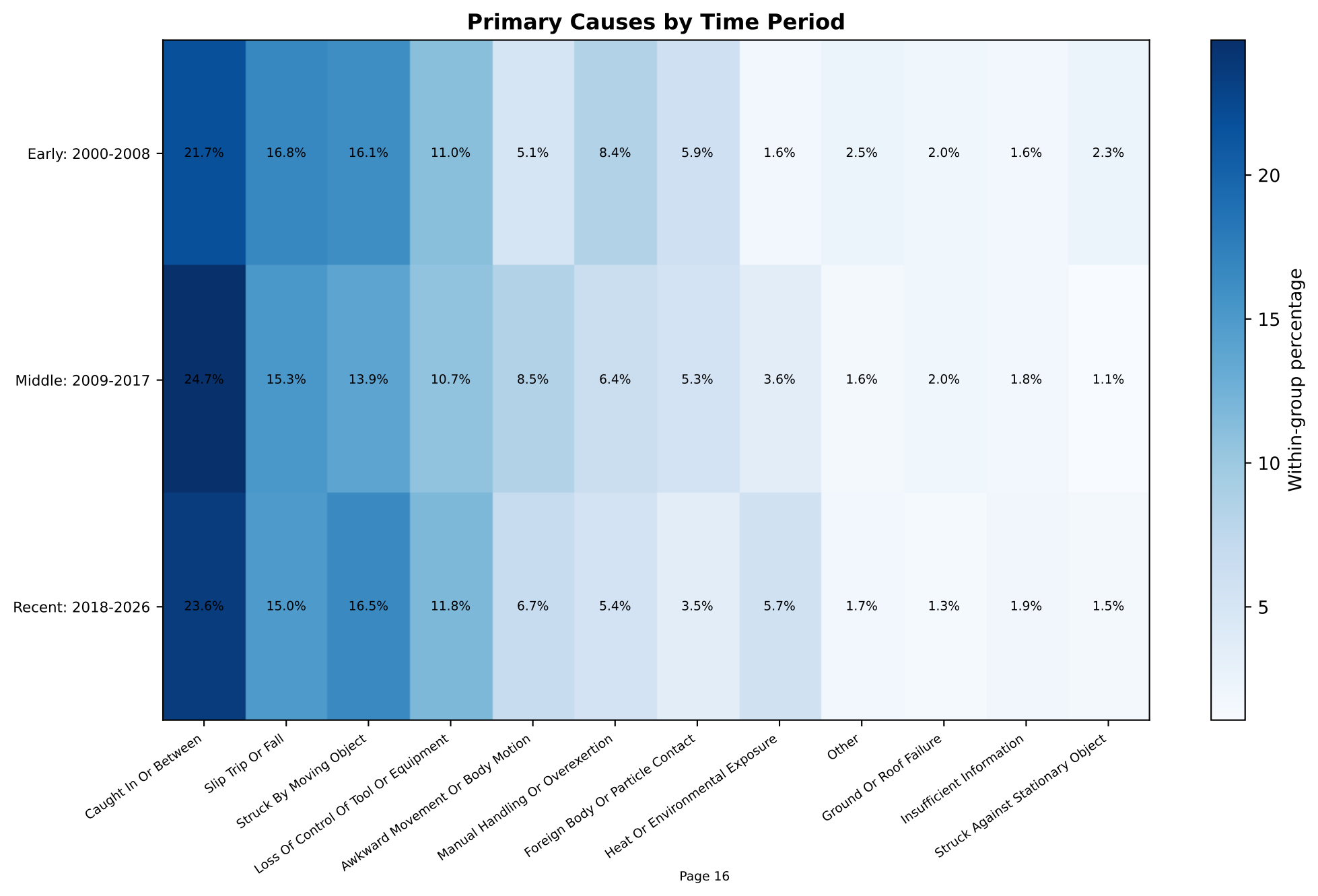
Prevention Sensitivity: Direct versus Expanded

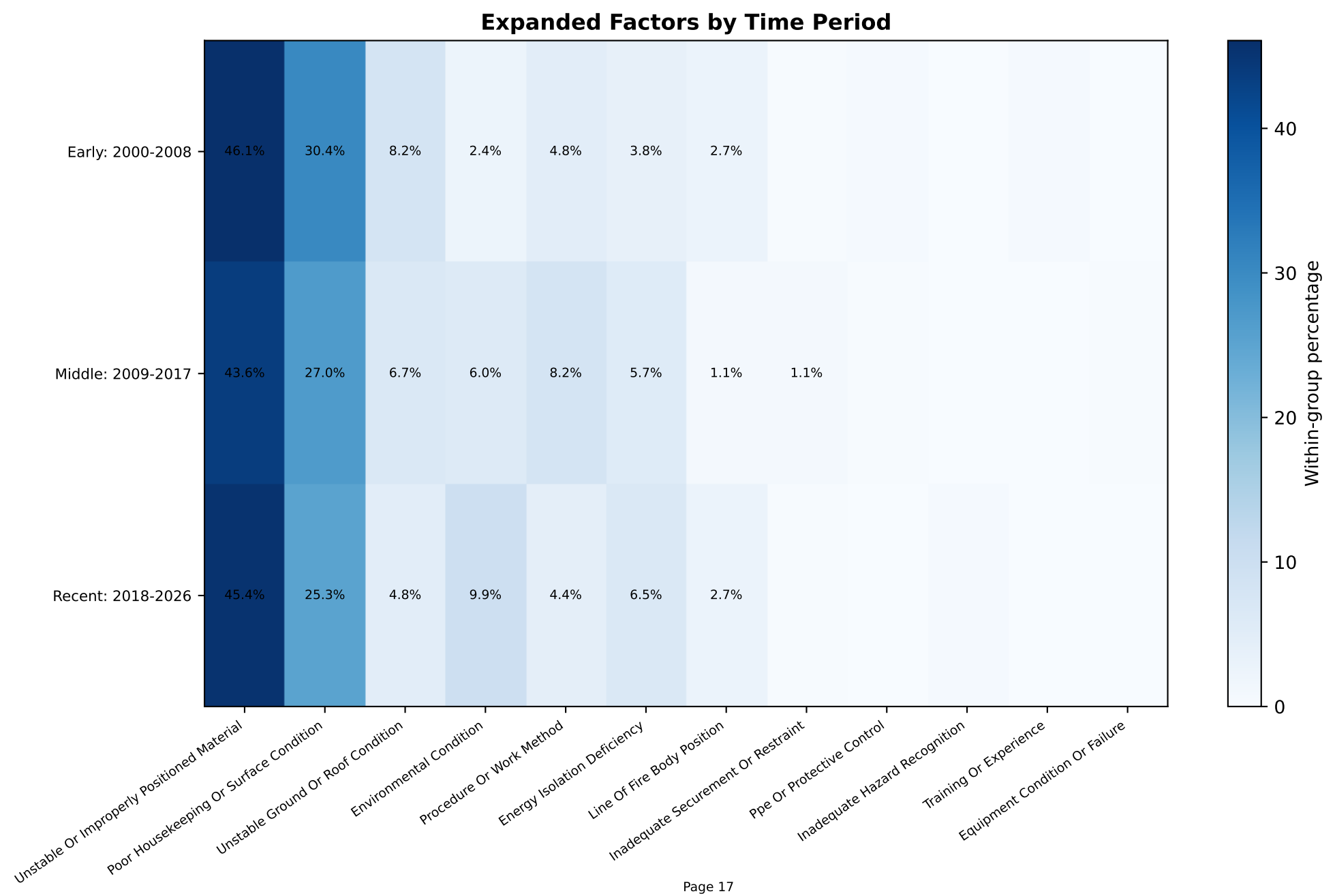
REVISED_PREVENTION	Direct support only	Direct plus contextual	Percentage-point difference
CONTROL_ENVIRONMENTAL_EXPOSURE	0.0000	6.1060	6.1060
IMPROVE_GROUND_CONTROL_OR_SCALING	4.9774	6.5668	1.5894
REVISE_PROCEDURE_OR_JOB_PLAN	4.9774	5.8756	0.8982
IMPROVE_PPE_OR_PROTECTIVE_EQUIPMENT	0.0000	0.3456	0.3456
CONDUCT_PRE_JOB_HAZARD_REVIEW	0.0000	0.3456	0.3456
ESTABLISH_LINE_OF_FIRE_EXCLUSION	2.2624	1.9585	-0.3039
REPAIR_REPLACE_OR_INSPECT_EQUIPMENT	0.4525	0.1152	-0.3373
ISOLATE_OR_CONTROL_STORED_ENERGY	5.8824	5.2995	-0.5828
PROVIDE_TARGETED_TRAINING	0.9050	0.2304	-0.6746
IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING	29.8643	27.5346	-2.3297
SECURE_OR_REPOSITION_MATERIAL	50.6787	45.6221	-5.0566

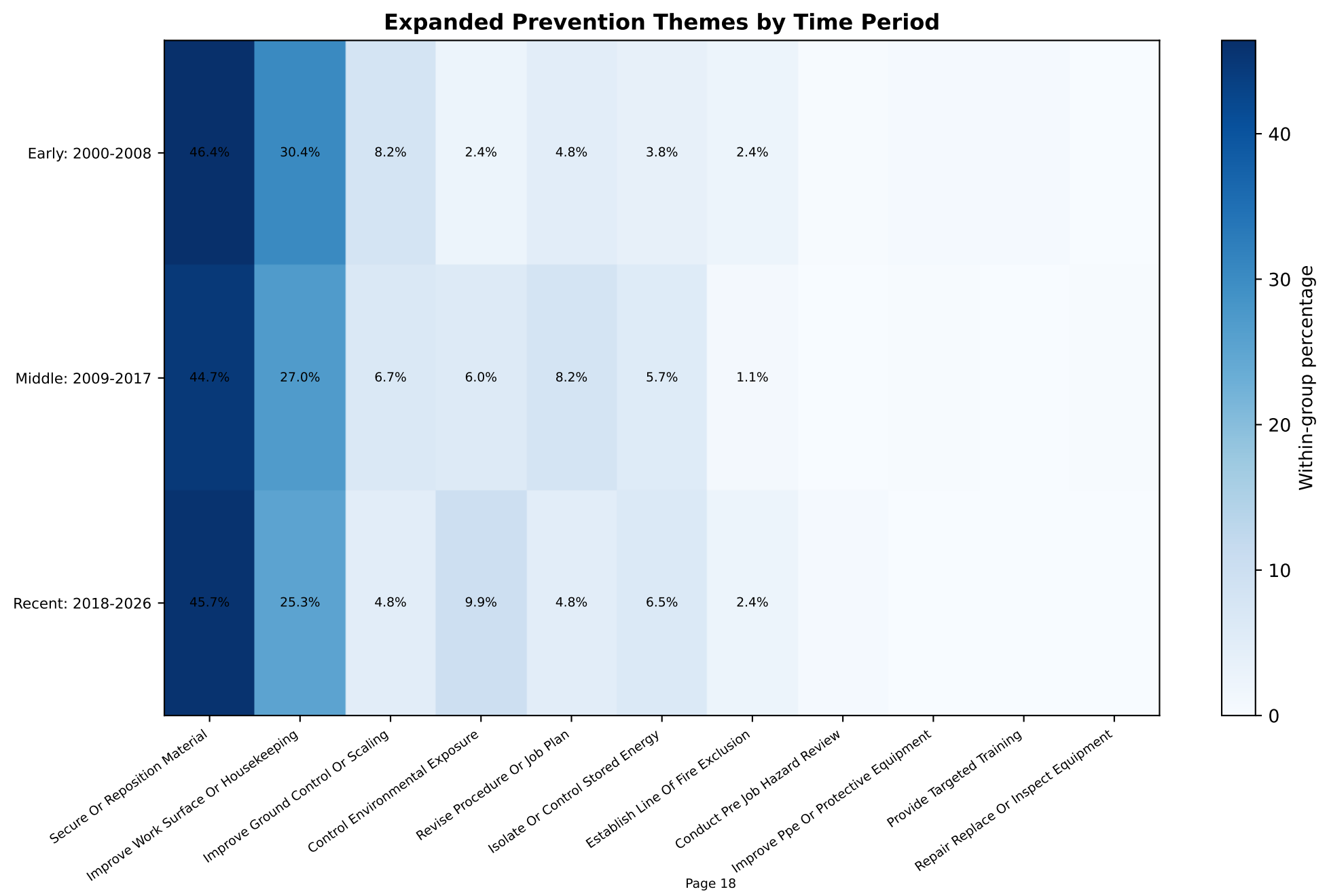

Primary Causes by Severity
Fatality
Permanent disability
Days away
Restricted or transferred
No lost time or other
29.7% 9.9% 18.1% 18.7% 6.2% 3.4% 1.1%
54.6% 3.6% 12.5% 10.6% 1.9% 10.3% 1.9% 1.7%
14.1% 21.5% 20.4% 9.0% 10.3% 6.8% 11.7% 2.2% 1.1%
11.4% 25.3% 13.6% 10.4% 15.4% 12.0% 4.5% 2.9%
4.9% 17.9% 12.7% 6.8% 4.9% 3.9% 6.2% 20.1% 5.5% 1.3% 5.5% 2.3%
Caught In Or Between
Slip Trip Or Fall
Struck By Moving Object
Loss Of Control Of Tool Or Equipment
Awkward Movement Or Body Motion
Manual Handling Or Overexertion
Foreign Body Or Particle Contact
Heat Or Environmental Exposure
Other
Ground Or Roof Failure
Insufficient Information
Struck Against Stationary Object
Within-group percentage
0
10
20
30
40
50


Expanded Factors by Severity
Fatality
Permanent disability
Days away
Restricted or transferred
No lost time or other
50.3%
13.7%
22.2%
10.5%
3.3%
77.8%
6.3%
2.6%
1.6%
5.3%
3.2%
1.6%
39.9%
41.9%
4.0%
9.1%
3.0%
1.0%
28.0%
47.4%
1.7%
12.6%
5.7%
2.3%
1.1%
25.5%
26.1%
4.6%
34.6%
4.6%
2.6%
1.3%
Unstable Or Improperly Positioned Material
Poor Housekeeping Or Surface Condition
Unstable Ground Or Roof Condition
Environmental Condition
Procedure Or Work Method
Energy Isolation Deficiency
Line Of Fire Body Position
Inadequate Securement Or Restraint
Ppe Or Protective Control
Inadequate Hazard Recognition
Training Or Experience
Equipment Condition Or Failure
Within-group percentage
0
10
20
30
40
50
60
70


Expanded Prevention Themes by Severity
Fatality
Permanent disability
Days away
Restricted or transferred
No lost time or other
50.3%
13.7%
22.2%
10.5%
3.3%
79.4%
6.3%
2.6%
2.1%
5.3%
2.6%
40.4%
41.9%
4.0%
9.1%
3.0%
28.6%
47.4%
1.7%
12.6%
5.7%
2.3%
1.1%
25.5%
26.1%
4.6%
34.6%
4.6%
2.6%
1.3%
Secure Or Reposition Material
Improve Work Surface Or Housekeeping
Improve Ground Control Or Scaling
Control Environmental Exposure
Revise Procedure Or Job Plan
Isolate Or Control Stored Energy
Establish Line Of Fire Exclusion
Conduct Pre Job Hazard Review
Improve Ppe Or Protective Equipment
Provide Targeted Training
Repair Replace Or Inspect Equipment
Within-group percentage
0
10
20
30
40
50
60
70


Primary Causes by Time Period
Early: 2000-2008
Middle: 2009-2017
Recent: 2018-2026
21.7%
16.8%
16.1%
11.0%
5.1%
8.4%
5.9%
1.6%
2.5%
2.0%
1.6%
2.3%
24.7%
15.3%
13.9%
10.7%
8.5%
6.4%
5.3%
3.6%
1.6%
2.0%
1.8%
1.1%
23.6%
15.0%
16.5%
11.8%
6.7%
5.4%
3.5%
5.7%
1.7%
1.3%
1.9%
1.5%
Caught In Or Between
Slip Trip Or Fall
Struck By Moving Object
Loss Of Control Of Tool Or Equipment
Awkward Movement Or Body Motion
Manual Handling Or Overexertion
Foreign Body Or Particle Contact
Heat Or Environmental Exposure
Other
Ground Or Roof Failure
Insufficient Information
Struck Against Stationary Object
Within-group percentage
20
15
10
5


Expanded Factors by Time Period
Early: 2000-2008
Middle: 2009-2017
Recent: 2018-2026
46.1%
30.4%
8.2%
2.4%
4.8%
3.8%
2.7%
43.6%
27.0%
6.7%
6.0%
8.2%
5.7%
1.1%
1.1%
45.4%
25.3%
4.8%
9.9%
4.4%
6.5%
2.7%
Unstable Or Improperly Positioned Material
Poor Housekeeping Or Surface Condition
Unstable Ground Or Roof Condition
Environmental Condition
Procedure Or Work Method
Energy Isolation Deficiency
Line Of Fire Body Position
Inadequate Securement Or Restraint
Ppe Or Protective Control
Inadequate Hazard Recognition
Training Or Experience
Equipment Condition Or Failure
Within-group percentage
0
10
20
30
40


Expanded Prevention Themes by Time Period
Early: 2000-2008
Middle: 2009-2017
Recent: 2018-2026
46.4%
30.4%
8.2%
2.4%
4.8%
3.8%
2.4%
44.7%
27.0%
6.7%
6.0%
8.2%
5.7%
1.1%
45.7%
25.3%
4.8%
9.9%
4.8%
6.5%
2.4%
Secure Or Reposition Material
Improve Work Surface Or Housekeeping
Improve Ground Control Or Scaling
Control Environmental Exposure
Revise Procedure Or Job Plan
Isolate Or Control Stored Energy
Establish Line Of Fire Exclusion
Conduct Pre Job Hazard Review
Improve Ppe Or Protective Equipment
Provide Targeted Training
Repair Replace Or Inspect Equipment
Within-group percentage
0
10
20
30
40

Early-to-Recent Cause Changes

Primary cause	Total count	Early percent	Recent percent	Percentage-point change
HEAT_OR_ENVIRONMENTAL_EXPOSURE	64	1.6474	5.7239	4.0765
CAUGHT_IN_OR_BETWEEN	411	21.7463	23.5690	1.8227
AWKWARD_MOVEMENT_OR_BODY_MOT	119	5.1071	6.7340	1.6269
CHEMICAL_DUST_OR_OCCUPATIONAL_E	28	1.3180	2.1886	0.8706
LOSS_OF_CONTROL_OF_TOOL_OR_EQU	197	11.0379	11.7845	0.7466
STRUCK_BY_MOVING_OBJECT	274	16.1450	16.4983	0.3533
INSUFFICIENT_INFORMATION	31	1.6474	1.8519	0.2044
PRESSURE_OR_STORED_ENERGY_RELE	22	0.8237	1.0101	0.1864
UNEXPECTED_EQUIPMENT_MOVEMENT	12	0.6590	0.6734	0.0144
GROUND_OR_ROOF_FAILURE	31	1.9769	1.3468	-0.6301
OTHER	34	2.4712	1.6835	-0.7877
STRUCK_AGAINST_STATIONARY_OBJECT	29	2.3064	1.5152	-0.7913
ELECTRICAL_CONTACT	18	1.4827	0.6734	-0.8093
SLIP_TRIP_OR_FALL	277	16.8040	14.9832	-1.8208
FOREIGN_BODY_OR_PARTICLE_CONTAC	87	5.9308	3.5354	-2.3955
MANUAL_HANDLING_OR_OVEREXERTIO	119	8.4020	5.3872	-3.0148

Early-to-Recent Factor Changes

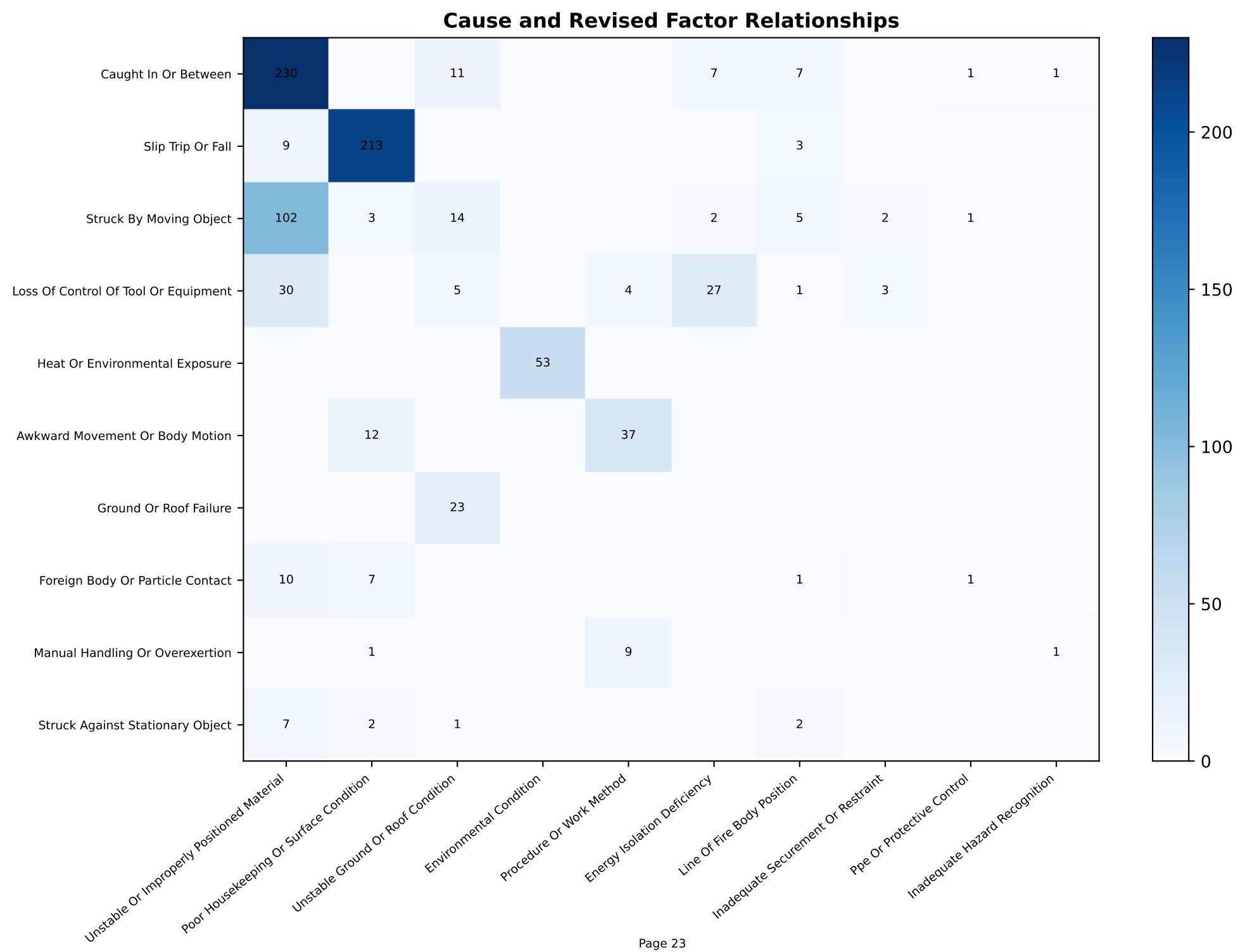
Contributing factor	Total count	Early percent	Recent percent	Percentage-point change
ENVIRONMENTAL_CONDITION	53	2.3891	9.8976	7.5085
ENERGY_ISOLATION_DEFICIENCY	46	3.7543	6.4846	2.7304
LINE_OF_FIRE_BODY_POSITION	19	2.7304	2.7304	0.0000
PROCEDURE_OR_WORK_METHOD	50	4.7782	4.4369	-0.3413
UNSTABLE_OR_IMPROPERLY_POSITIONE	391	46.0751	45.3925	-0.6826
UNSTABLE_GROUND_OR_ROOF_CONDI	57	8.1911	4.7782	-3.4130
POOR_HOUSEKEEPING_OR_SURFACE_C	239	30.3754	25.2560	-5.1195

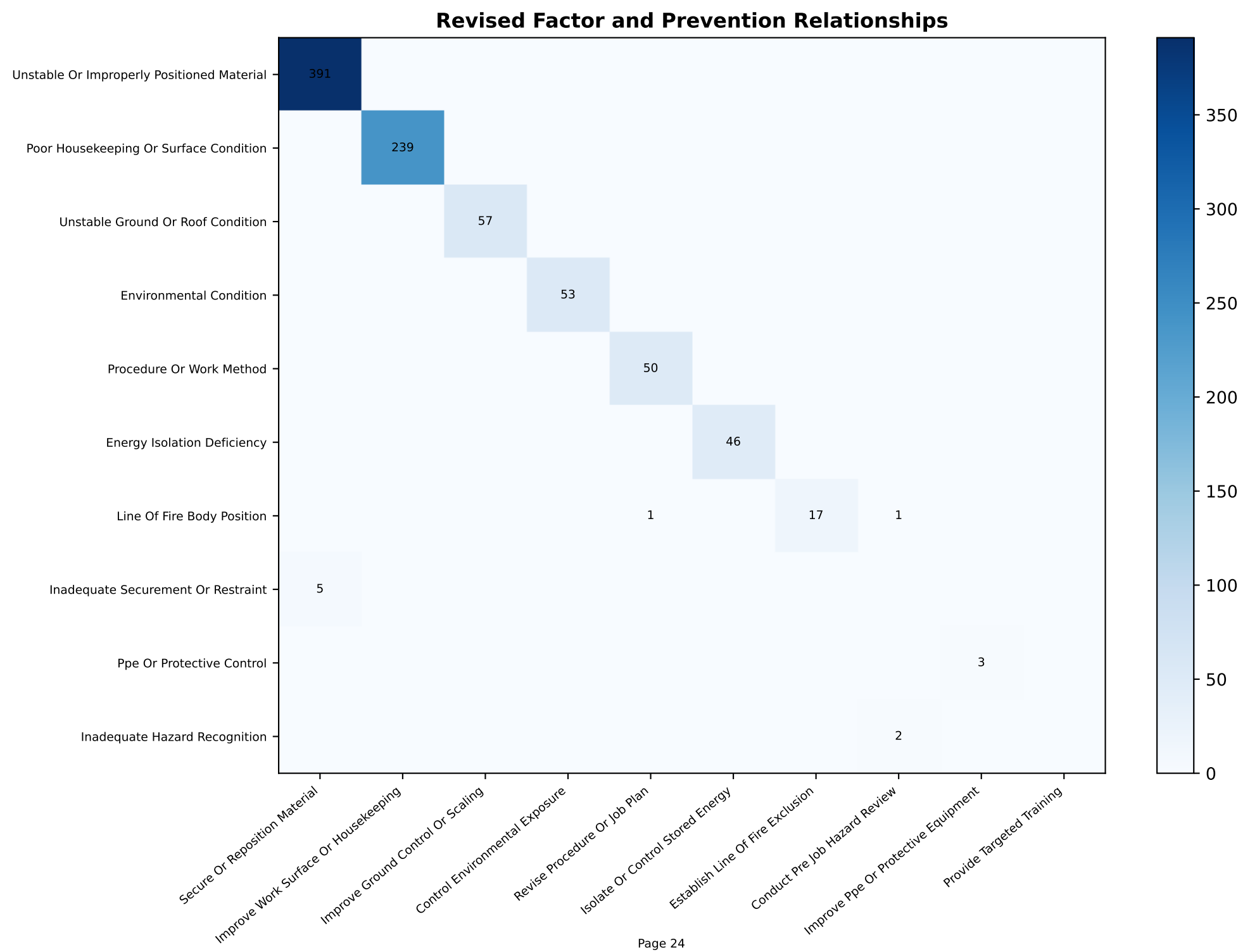
Early-to-Recent Prevention Changes

Prevention theme	Total count	Early percent	Recent percent	Percentage-point change
CONTROL_ENVIRONMENTAL_EXPOSURE	53	2.3891	9.8976	7.5085
ISOLATE_OR_CONTROL_STORED_ENER	46	3.7543	6.4846	2.7304
ESTABLISH_LINE_OF_FIRE_EXCLUSION	17	2.3891	2.3891	0.0000
REVISE_PROCEDURE_OR_JOB_PLAN	51	4.7782	4.7782	0.0000
SECURE_OR_REPOSITION_MATERIAL	396	46.4164	45.7338	-0.6826
IMPROVE_GROUND_CONTROL_OR_SCA	57	8.1911	4.7782	-3.4130
IMPROVE_WORK_SURFACE_OR_HOUSEK	239	30.3754	25.2560	-5.1195

Association Tests

Comparison	Chi-square	Degrees of freedom	P-value	Cramer's V	Records
Cause by severity	1104.7134	72	0.0000	0.3957	1764
Factor by severity	540.4395	44	0.0000	0.3945	868
Prevention by severity	533.8583	40	0.0000	0.3921	868
Cause by time	48.6926	36	0.0770	0.1175	1764
Factor by time	40.2061	22	0.0102	0.1522	868
Prevention by time	35.1822	20	0.0192	0.1424	868


Cause and Revised Factor Relationships
Caught In Or Between
Slip Trip Or Fall
Struck By Moving Object
Loss Of Control Of Tool Or Equipment
Heat Or Environmental Exposure
Awkward Movement Or Body Motion
Ground Or Roof Failure
Foreign Body Or Particle Contact
Manual Handling Or Overexertion
Struck Against Stationary Object
230 11 7 7 1 1
9 213 3
102 3 14 2 5 2 1
30 5 4 27 1 3
53
12 37
23
10 7 1 1
1 9 1
7 2 1 2
Unstable Or Improperly Positioned Material
Poor Housekeeping Or Surface Condition
Unstable Ground Or Roof Condition
Environmental Condition
Procedure Or Work Method
Energy Isolation Deficiency
Line Of Fire Body Position
Inadequate Securement Or Restraint
Ppe Or Protective Control
Inadequate Hazard Recognition
200
150
100
50
0


Revised Factor and Prevention Relationships
Unstable Or Improperly Positioned Material
Poor Housekeeping Or Surface Condition
Unstable Ground Or Roof Condition
Environmental Condition
Procedure Or Work Method
Energy Isolation Deficiency
Line Of Fire Body Position
Inadequate Securement Or Restraint
Ppe Or Protective Control
Inadequate Hazard Recognition
391
239
57
53
50
46
1
17
1
5
3
2
Secure Or Reposition Material
Improve Work Surface Or Housekeeping
Improve Ground Control Or Scaling
Control Environmental Exposure
Revise Procedure Or Job Plan
Isolate Or Control Stored Energy
Establish Line Of Fire Exclusion
Conduct Pre Job Hazard Review
Improve Ppe Or Protective Equipment
Provide Targeted Training
350
300
250
200
150
100
50
0

Interpretation and Limitations

Cause results use valid acute accident mechanisms.

Expanded factor and prevention results include direct and contextual evidence tiers. They should be compared with the strict direct-support subset and validated in Blocks 3 and 4.

Category-specific phrase rules are transparent but imperfect. They can recover plausible factors missed by verbatim matching, but they may also admit false positives.

Time-period changes are narrative-share changes, not exposure-adjusted incidence rates.

The sample is balanced by severity and does not represent the natural severity distribution.